

Synopsis
on
Agentic Deployment Kit

Project Based Learning -V (22AI014)

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Introduction

In today's software industry, developers are expected to not only write code but also manage complex deployment workflows using tools like AWS, Docker, Kubernetes, and CI/CD pipelines. This can be overwhelming—especially for those who are just getting started or come from non-cloud backgrounds. Many developers face a steep learning curve, spending more time figuring out infrastructure and DevOps tools than focusing on building software.

This report introduces an innovative solution: an autonomous AI agent for software deployment. The goal is to reduce the burden on developers by allowing an intelligent system to handle most of the deployment tasks automatically. Instead of learning every detail of cloud platforms and configuration files, developers will be able to focus on what matters most—building features—while the AI handles the rest.

Traditional CI/CD systems are helpful but often require manual setup, break easily, and can be hard to manage across different projects and environments. Our proposed system uses AI to understand, plan, and continuously improve the deployment process—making it more reliable, faster, and less stressful for developers.

Problem Formulation

In our research we discovered that many developers, regardless of experience level, face serious frustrations with current deployment workflows. These problems go beyond just tooling—they affect productivity, learning, and even confidence. Here are the major issues that need to be solved:

1. Manual and Risky Deployment Practices

Many developers still deploy by manually copying files or scripts to production servers. This process is slow, error-prone, and highly stressful. A small mistake can bring down an entire application. Despite knowing it's not ideal, they continue with this method due to a lack of better, accessible alternatives.

2. Inflexible Deployment Tools

Existing tools and extensions often don't support the actual structure or needs of real-world projects. For example, they may assume a standard project layout and break when asked to deploy from a custom folder like `/src`. This makes automation harder than it should be, leaving developers stuck with "almost-working" solutions.

3. Lack of Clear Guidance Creates Anxiety

Many developers are unsure how deployment is "supposed" to work. Online tutorials are either too basic or too complex, leading to confusion. This knowledge gap causes many to feel underprepared—especially in interviews—leading to professional anxiety and self-doubt.

4. Environment-Specific Failures ("It works on my machine")

A common issue is that applications work perfectly on local setups but fail in staging or production. This is usually due to configuration mismatches or unsupported tech stacks. Debugging such problems is difficult and time-consuming, especially when documentation is lacking.

5. Deployment Is a Distraction from Core Work

Developers in roles like data science, machine learning, or app development often find deployment tasks to be a distraction from their actual work. They spend valuable time solving DevOps problems instead of focusing on what they were hired to do—building models, writing features, or analyzing data.

Proposed Solution

To simplify the entire deployment process, we are building a **VS Code extension** that acts like an intelligent DevOps assistant. It will help developers deploy their projects without deep knowledge of AWS, Docker, or CI/CD pipelines. Everything will be automated—from understanding your codebase to launching your app in production.

Here's what it will do:

- **Auto-Detect Stack & Structure**

On opening a project, the extension will automatically detect the tech stack (e.g., Node.js, Python, React) and understand the folder structure (e.g., where the entry point and environment files are).

- **Generate Docker & Compose Files**

It will generate a **Dockerfile** and **docker-compose.yml** customized to your project—no need to write or understand Docker syntax.

- **Securely Handle AWS Credentials**

The extension will collect your AWS keys using a secure form, encrypt them locally, and use them only during deployment. Credentials won't be hardcoded or exposed.

- **Ask Deployment Questions via Chat**

A built-in chatbot will ask simple questions like:

- “What specs do you want for the server (CPU/RAM)?”
- “Do you want HTTPS or a custom domain?”
- “Should this service be public or private?”

Based on answers, it configures the infrastructure.

- **No DevOps Knowledge Required**

Even if the developer doesn't know AWS, Docker, or YAML, the chatbot will guide them step-by-step and auto-generate everything needed.

- **Fixes Errors Automatically**

If a deployment fails, it will analyze the logs, detect the issue, and either suggest or auto-fix it—reducing debugging time drastically.

Flowchart



